

SYLLABUS

UNIT I: Introduction to Machine Learning

Introduction to Machine Learning, Learning Paradigms, Machine Learning Process, PAC Learning, Version Spaces, Role of Machine Learning in Artificial Intelligence Applications, Perceptron-Single Layer, Multilayer, Metrics-Accuracy, Precision, Recall, F1-Score.

UNIT II: Supervised Learning

Regression: Linear Regression, Multiple Linear Regression, Logistic Regression.

Classification: Types of Classification-Binary, Multiclass, Multilabel, Naïve Bayes Classifier, Decision Tree, ID3, CART, KNN Classifier, Support Vector Machine, Error Bounds.

UNIT III: Unsupervised Learning

Clustering Basics-Partitioned, Hierarchical and Density Based, K-Means Clustering, K-Mode Clustering, Self-Organizing Maps.

Dimensionality Reduction: Principal Component Analysis, Kernel PCA, t-SNE (t-distributed Stochastic Neighbor Embedding), Metrics & Error Correction.

UNIT IV: Reinforcement Learning

Basics of RL, RL Framework, Markov Decision Process, Exploration Vs Exploitation, Policies, Value Functions and Bellman Equations, Solution Methods, Q-Learning, Uses of Reinforcement Learning.

UNIT V: Ensemble Learning

Bagging and Boosting-AdaBoost, XGBoost inclusive and Stumping, Random Forests, Comparison of Random Forest with Boosting, Metrics & Error Correction.

UNIT VI: Model Evaluation and Selection

Cross Validation, Bias-Variance Tradeoff, Class Imbalance: SMOTE, One Class SVM, Optimization of Hyperparameters.